



Modelling Financial Markets

Seminar 1: Stylised Facts and Returns — Hands-on



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Acronyms used in this material

- ACF: AutoCorrelation Function
- AI: Artificial Intelligence
- ARCH-LM: ARCH Lagrange Multiplier (test)
- BET: Bucharest Exchange Trading (index)
- BET-TR: BET Total Return (index)
- BNR: Banca Națională a României (the National Bank of Romania)
- CAGR: Compound Annual Growth Rate
- COVID-19: COronaVirus Disease 2019
- EGARCH: Exponential GARCH
- ETF: Exchange-Traded Fund
- FTSE: Financial Times Stock Exchange (FTSE Russell, index provider)
- GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- GJR-GARCH: Glosten–Jagannathan–Runkle GARCH
- HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- HAR: Heterogeneous AutoRegressive (model)
- JB: Jarque–Bera (test)
- MLE: Maximum Likelihood Estimation
- OHLC: Open, High, Low, Close
- SE: Standard Error

Outline

Introduction

Key Definitions

Part A: Pen-and-Paper Exercises

Part B: Verify (B1–B8)

Part B: Test (B9–B14)

Part C: Open Question

Wrap-up

Seminar Objectives

By the end of this seminar you can

- Compute and interpret the statistics behind the stylised facts
- Derive where heavy tails and volatility clustering come from (mixtures, regimes, GARCH)
- Estimate tail indices and decide which moments exist
- Test autocorrelation, asymmetry and performance differences with inference robust to heavy tails and clustering
- Check which facts a model reproduces

□ Background: Lecture 1 and [Cont \(2001\)](#); textbook: [Franke, Härdle & Hafner \(2019\)](#)

Structure

- **Part A:** nine pen-and-paper exercises, from warm-up (A1–A6) to derivations (A7–A9)
- **Part B:** fourteen computational exercises: verify (B1–B8), then test (B9–B14)
- **Part C:** one open question, a starting point for the team project (C1), and an AI answer to critique (C2)

How to Work in This Seminar

- **[Solved]** problems show the full worked solution; **[Proposed]** problems show only the task: their solutions are discussed in class
- Every Part B exercise ends with **interpretation questions**
 - ▶ They cannot be answered by reading a number: ask what would change the conclusion
- Report every estimate with its uncertainty (standard error, interval or p-value)
- Before any test: look at the data (charts, extremes, zero returns, calendar)

Solved and Proposed Problems

Solved: worked examples in this deck

- Part A: A1 (aggregating returns), A3 (Jarque–Bera and Ljung–Box, step by step), A5 (Parkinson volatility), A6 (kurtosis of a mixture), A7 (the Hill estimator, step by step)
- Part B: B1 (descriptive statistics), B3 (uncorrelated but not independent), B9 (robust Ljung–Box), B11 (Hill on both tails)

Proposed: you solve them

- Part A: A2, A4 (model: A1, A5); A8, A9 (model: A6)
- Part B: B2, B6, B7 (model: B1); B4, B5 (model: B3); B8 (model: A5)
- Part B, inference: B10, B12, B13 (model: B9); B14 (model: B3, B11)
- Part C: C1, open question and project seed; C2, critique an AI answer (model: A3, B3, B9)

Setup: Python Environment and Data

-  **Open in Colab:** [chapter1_seminar_notebook.ipynb](#)
- Google Colab (recommended) or local Python ≥ 3.10 with `numpy`, `pandas`, `scipy`, `statsmodels`, `arch`, `matplotlib`
- Daily market data are in the course repository, folder `data/market`

```
import numpy as np, pandas as pd
URL = "https://raw.githubusercontent.com/danpele/MFM/main/data/market/"
bet = pd.read_csv(URL + "BETTR.INDX.csv", index_col="date", parse_dates=True)
r = np.log(bet["close"]).diff().dropna()           # daily log returns
```

- Other files: `BTC-USD.CC.csv`, `GSPC.INDX.csv`; EUR/RON: BNR official reference rate
- Joint analyses (correlations, Sharpe comparisons): join the *prices* on common days first, then take returns

Key Definitions: Returns and Annualisation

- ▣ **Simple return** $R_t = P_t/P_{t-1} - 1$; **log return** $r_t = \ln(1 + R_t) = \ln P_t - \ln P_{t-1}$
 - ▶ Log returns add over time; simple returns add across assets
- ▣ **Portfolio return** $R_{p,t} = \sum_i w_i R_{i,t}$, with weights w_i summing to 1
- ▣ **Annualisation** with P periods per year (252 for stocks, 365 for crypto)
 - ▶ $\mu_{\text{ann}} = P \bar{r}$ and $\sigma_{\text{ann}} = \sqrt{P} \hat{\sigma}$, valid for i.i.d. (independent and identically distributed) returns
- ▣ **CAGR** (Compound Annual Growth Rate) $= (P_T/P_0)^{1/Y} - 1$, with Y the number of years
- ▣ **h -day return**: sum of h consecutive daily log returns; non-overlapping blocks share no days

Key Definitions: Moments and Tests

- ▣ **Skewness** $\hat{S} = \frac{1}{T} \sum_t z_t^3$ and **kurtosis** $\hat{K} = \frac{1}{T} \sum_t z_t^4$, with $z_t = (r_t - \bar{r})/\hat{\sigma}$
 - ▶ Normal distribution: $S = 0$, $K = 3$; **excess kurtosis** $= K - 3$; for i.i.d. data from the Normal distribution $SE(\hat{K} - 3) \approx \sqrt{24/T}$
- ▣ **Jarque–Bera** (JB) normality test: $JB = \frac{T}{6} (\hat{S}^2 + (\hat{K} - 3)^2/4) \sim \chi_2^2$
- ▣ **Autocorrelation** $\hat{\rho}_k = \text{corr}(r_t, r_{t-k})$; **ACF** (AutoCorrelation Function): $\hat{\rho}_k$ against k
- ▣ **Ljung–Box** test: $Q(m) = T(T+2) \sum_{k=1}^m \hat{\rho}_k^2 / (T-k) \sim \chi_m^2$ under i.i.d. returns
- ▣ **ARCH-LM** test (AutoRegressive Conditional Heteroskedasticity, Lagrange Multiplier): TR^2 from regressing $\hat{\varepsilon}_t^2$ on q lags, $\sim \chi_q^2$
- ▣ **p-value**: probability under H_0 of a statistic at least as extreme as observed; reject at 5% if below 0.05
- ▣ **Student-t** with ν degrees of freedom: heavy tails for small ν , the Normal distribution as $\nu \rightarrow \infty$

Key Definitions: Risk and Performance

- ▣ **VaR** (Value-at-Risk) at 1%, one day: the loss exceeded on 1% of days, $\text{VaR} = -q_{0.01}$
 - ▶ q_α : quantile, the value below which a fraction α of returns falls; a **breach**: a day whose loss is larger than the VaR
- ▣ **Drawdown** $DD_t = P_t / \max_{s \leq t} P_s - 1$; **maximum drawdown** $MDD = \min_t DD_t$
- ▣ **Sharpe ratio** $SR = (\mu_{\text{ann}} - r_f) / \sigma_{\text{ann}}$, r_f the risk-free rate (here 0)
- ▣ **Sortino ratio**: as Sharpe, with the downside deviation $\sigma^- = \sqrt{\mathbb{E}[\min(r_t, 0)^2]}$ in the denominator
- ▣ **Calmar ratio** = $\text{CAGR} / |MDD|$
- ▣ **Leverage correlation** $L(k) = \text{corr}(r_t, |r_{t+k}|)$: negative when losses raise future volatility

Key Definitions: Volatility Estimators and Data

- ▣ **OHLC**: Open, High, Low and Close prices of a day; o_t, h_t, l_t, c_t their logs
- ▣ **Close-to-close** volatility: standard deviation of daily log returns
- ▣ **Range-based** estimators (formulas in Lecture 1)
 - ▶ Parkinson: daily range $h_t - l_t$, assumes **zero drift**; Garman–Klass: range plus open-to-close move
 - ▶ Rogers–Satchell: **drift-independent**; Yang–Zhang: adds the overnight jump $o_t - c_{t-1}$
- ▣ **BET-TR**: Total Return version of the Bucharest Stock Exchange index BET, dividends re-invested
- ▣ **EUR/RON**: BNR reference rate; a **managed float**, i.e. a market rate with central-bank interventions

Key Definitions: Inference Tools

- ▣ **Hill estimator** [Hill \(1975\)](#) from the k largest values $X_{(1)} \geq \dots \geq X_{(k+1)}$
 - ▶ $1/\hat{\alpha} = \frac{1}{k} \sum_{i=1}^k \ln(X_{(i)}/X_{(k+1)})$, $\text{SE}(\hat{\alpha}) \approx \hat{\alpha}/\sqrt{k}$; $\mathbb{E}|X|^p$ exists for $p < \alpha$
- ▣ **Robust autocorrelation variance**, $x_t = r_t - \bar{r}$: $\hat{\tau}_k = \sum_t x_t^2 x_{t-k}^2 / (\sum_t x_t^2)^2$ (i.i.d.: $\hat{\tau}_k \approx 1/T$)
 - ▶ Robust portmanteau $\tilde{Q}(m) = \sum_{k=1}^m \hat{\rho}_k^2 / \hat{\tau}_k \sim \chi_m^2$ [Lobato, Nankervis & Savin \(2001\)](#)
- ▣ **HAC** (Heteroskedasticity and Autocorrelation Consistent) standard errors [Newey & West \(1987\)](#): valid under clustering
- ▣ **Block bootstrap**: resample blocks of consecutive days, keeping the dependence inside each block
- ▣ **Non-synchronous trading**: markets closing at different hours record the same news on different days [Scholes & Williams \(1977\)](#), [Dimson \(1979\)](#)

Exercise A1: Aggregating Returns over Time and across Assets [Solved]

Task

- A share closes at 100, 110, 99, 104.5: compute the daily R_t and r_t and the 3-day return in both ways
- Is $\sum R_t$ the 3-day simple return? Is $\sum r_t$ the 3-day log return?
- Portfolio 60% A, 40% B with $R_A = +5\%$, $R_B = -2\%$: compare R_p , the weighted log return and $\ln(1 + R_p)$

Solution

- $R = (+10\%, -10\%, +5.56\%)$,
 $r = (0.0953, -0.1054, 0.0541)$
- 3-day: 4.5% but $\sum R_t = 5.56\%$;
 $\ln 1.045 = 0.0440 = \sum r_t$
- $R_p = 2.20\%$; weighted log 0.02119 vs true 0.02176
- Over time use log returns; across assets use simple returns

Exercise A2: Annualisation [Proposed]

Solution discussed in class

Task

- ▣ An equity index has daily mean log return 0.04% and daily volatility 1.2%
- ▣ Annualise mean and volatility (252 days); compute the monthly (21-day) volatility
- ▣ Bitcoin has daily volatility 3.5%: annualise it; which P do you use?
- ▣ Under which assumption is the $\sqrt{P}$ rule valid?

Exercise A3: Jarque–Bera and Ljung–Box, Step by Step [Solved]

Task

- ▣ $T = 2500$ returns with $\hat{S} = -0.5$, $\hat{K} = 8$: compute JB Jarque & Bera (1980) and decide at 5% (χ^2_2 : 5.99)
- ▣ $T = 1000$, $\hat{\rho}_1 = 0.10$, $\hat{\rho}_2 = -0.05$, $\hat{\rho}_3 = 0.03$: compute $Q(3)$ Ljung & Box (1978) and decide (χ^2_3 : 7.81)
- ▣ Both tests assume i.i.d. data under H_0 : which stylised fact makes $Q(3)$ too large?

Solution

- ▣ $JB = \frac{2500}{6}(0.25 + 6.25) = 2708.3$: reject; the kurtosis term dominates
- ▣ $Q(3) = 1000 \cdot 1002 \left(\frac{0.01}{999} + \frac{0.0025}{998} + \frac{0.0009}{997} \right) = 13.44$: reject
- ▣ Volatility clustering: $\text{Var}(\hat{\rho}_k) > 1/T$, so Q over-rejects (Exercise B9)

Exercise A4: Drawdown and Calmar Ratio [Proposed]

Solution discussed in class

Task

- ▣ Portfolio values at year-end: 100, 120, 90, 110, 130, 117
- ▣ Compute the drawdown series and the maximum drawdown
- ▣ With a 5-year CAGR of 3.19%, compute the Calmar ratio
- ▣ Why can two strategies with equal Sharpe ratios have very different Calmar ratios?

Exercise A5: Parkinson Volatility, Step by Step [Solved]

Task

- Five days with high/low ratios 1.020, 1.015, 1.030, 1.010, 1.025
- Compute $\hat{\sigma}_P^2 = \frac{1}{4 \ln 2} \overline{(\ln H - \ln L)^2}$ Parkinson (1980) and annualise (252 days)
- The estimator assumes a driftless Brownian motion: what happens with a strong trend? Which information does it not use at all?

Solution

- Mean of squares 0.000439; $\hat{\sigma}_P = 1.26\%$ per day, 20.0% per year
- Zero drift assumed: a trend widens the range and biases $\hat{\sigma}_P$ upward; Rogers & Satchell (1991) is drift-independent
- It does not use the overnight jump (close to next open): Yang & Zhang (2000) adds it

Exercise A6: Kurtosis of a Volatility Mixture [Solved]

Task

- ▣ $r_t = \sigma_t \varepsilon_t$, $\varepsilon_t \sim N(0, 1)$ independent of σ_t
- ▣ $\sigma_t = 1\%$ with probability 0.8 and 2% with probability 0.2
- ▣ Show that $K = 3\mathbb{E}[\sigma^4]/(\mathbb{E}[\sigma^2])^2$ and compute it
- ▣ Is the unconditional distribution the Normal distribution? Link to [Clark \(1973\)](#)

Solution

- ▣ $\mathbb{E}[r^4] = \mathbb{E}[\sigma^4]\mathbb{E}[\varepsilon^4] = 3\mathbb{E}[\sigma^4]$, $\mathbb{E}[r^2] = \mathbb{E}[\sigma^2]$
- ▣ In %²: $\mathbb{E}[\sigma^2] = 1.6$, $\mathbb{E}[\sigma^4] = 4.0$,
 $K = 12/2.56 = 4.69$
- ▣ Within each regime returns follow the Normal distribution, but the mixture is leptokurtic: heavy tails from random volatility

Exercise A7: The Hill Estimator, Step by Step [Solved]

Task

- ▣ The 11 largest absolute returns (%) of a hypothetical sample
 - ▶ 12.0; 9.5; 8.2; 7.4; 6.9; 6.1; 5.8; 5.5; 5.2; 5.0; 4.8
- ▣ Use the Hill estimator of the Key Definitions Hill (1975)
- ▣ (a) Compute $\hat{\alpha}$ for $k = 10$
- ▣ (b) Standard error $\hat{\alpha}/\sqrt{k}$ and the 95% interval
- ▣ (c) Which moments exist for the point estimate? What can you claim with 95% confidence?
- ▣ (d) Why not choose a very large k to reduce the standard error?

Solution

- ▣ (a) Log-ratios to 4.8: 0.916; 0.683; 0.536; 0.433; 0.363; 0.240; 0.189; 0.136; 0.080; 0.041
 - ▶ Sum 3.616, mean 0.3616, so $\hat{\alpha} = 2.77$
- ▣ (b) $SE = 2.77/\sqrt{10} = 0.87$; interval [1.05; 4.48]
- ▣ (c) Point estimate: variance exists, fourth moment does not; but the interval contains values below 2 and above 4, so with $k = 10$ we cannot decide either
- ▣ (d) A large k uses observations from the body, where the power law fails: less variance, more bias; read the Hill plot where it is flat
- ▣ On the 11 largest BET-TR days (2014–2026): $\hat{\alpha}_{10} = 3.38$ (SE 1.07)

Exercise A8: Where the Autocorrelation of r_t^2 Comes From [Proposed]

Solution discussed in class

Task

- ▣ As in A6: $r_t = \sigma_t \varepsilon_t$, $\sigma_t \in \{1\%, 2\%\}$ with probabilities 0.8 and 0.2
- ▣ Now the regime is a Markov chain [Hamilton \(1989\)](#): $P(\text{high} \rightarrow \text{high}) = 0.9$
- ▣ (a) Find $P(\text{low} \rightarrow \text{high})$ such that the stationary distribution stays (0.8; 0.2)
- ▣ (b) Show that $\text{Cov}(r_t^2, r_{t-k}^2) = \text{Var}(\sigma_t^2) \lambda^k$, with $\lambda = p_{LL} + p_{HH} - 1$
- ▣ (c) Compute $\text{corr}(r_t^2, r_{t-1}^2)$ and at lag 10
- ▣ (d) For the S&P 500 the ACF of r_t^2 is about 0.3 at lag 1 and still positive at lag 100 (Chapter 1): what does the model fail to do?

Exercise A9: Can a Gaussian GARCH Produce the S&P 500 Tails? [Proposed]

Solution discussed in class

Task

- ▣ GARCH(1,1) with $N(0, 1)$ innovations: $\sigma_t^2 = \omega + \alpha r_{t-1}^2 + \beta \sigma_{t-1}^2$ [Bollerslev \(1986\)](#); full model in Chapter 5
- ▣ Unconditional kurtosis: $K = 3[1 - (\alpha + \beta)^2] / [1 - (\alpha + \beta)^2 - 2\alpha^2]$
- ▣ (a) Compute K for $(\alpha, \beta) = (0.10; 0.85)$ and $(0.10; 0.88)$
- ▣ (b) Which condition gives a finite fourth moment?
- ▣ (c) With $\alpha + \beta = 0.99$, which α gives $K = 13.9$ (the S&P 500 excess of 10.9)? Where is this point relative to (b)?
- ▣ (d) What do you conclude about the leptokurtosis of standardised residuals?

Exercise B1: Descriptive Statistics [Solved]

Task

- Load BET-TR, Bitcoin and EUR/RON; compute daily log returns
- Report mean, standard deviation, annualised volatility, skewness, excess kurtosis, min, max; run Jarque–Bera

Interpretation

- Recompute the BET-TR kurtosis without its single largest day: what would change the conclusion that the tails are heavy?

Results

	BET-TR	Bitcoin	EUR/RON
<i>N</i>	2992	4384	5352
Ann. vol. (%)	15.5	66.6	4.4
Skewness	−1.40	−0.70	0.73
Excess kurtosis	17.8	12.0	15.1
Min log return (%)	−11.9	−46.5	−2.5
JB	40433	26482	51575

- Without 19 Dec 2018: 17.8 → 11.8; without the 5 largest days: 5.95
- Kurtosis is fragile, the tails remain heavy: B10

Exercise B2: Heavy Tails and 1% VaR on BET-TR [Proposed]

Solution discussed in class

Task

- Standardise BET-TR returns; count days beyond 3σ ; fit a Student- t by maximum likelihood (MLE)
- Compute the 1% one-day VaR under the Normal distribution, Student- t and the empirical quantile; count the breaches
- All three are estimated and checked on the **same** sample: a descriptive, in-sample comparison, not a backtest

Interpretation

- Why does the empirical VaR have about 30 breaches by construction? How would you turn this into a genuine backtest (Chapter 8)?

Exercise B3: Uncorrelated but not Independent (Bitcoin) [Solved]

Task

- Plot the ACF of r_t and $|r_t|$ for Bitcoin (lags 1–100)
- Compute Ljung–Box $Q(10)$ on r_t and on r_t^2 ; run ARCH-LM with 5 lags [Engle \(1982\)](#)

Interpretation

- The p-value of $Q(10)$ on r_t is about 3%: is that evidence of predictability, given what $Q(10)$ on r_t^2 says about the i.i.d. assumption (B9)?

Results

- Lag-1 autocorrelation: -0.02 for r_t , 0.21 for $|r_t|$
- $Q(10)$ on r_t : 20.2 ($p = 0.028$); on r_t^2 : 212.7 ; ARCH-LM: 114.7
- Strong clustering violates the i.i.d. null of $Q(10)$: the robust $\tilde{Q}(10)$ gives $p = 0.29$
- Direction unpredictable, size predictable

Exercise B4: Aggregational Gaussianity (Bitcoin) [Proposed]

Solution discussed in class

Task

- Build non-overlapping h -day log returns for $h = 1, 5, 20$ and compute excess kurtosis with the approximate SE $\sqrt{24/N}$
- Why should overlapping windows **not** be used here?

Interpretation

- $h = 60$ is a quarter for stocks but two months for Bitcoin: how would you compare horizons across markets? Why is $\sqrt{24/N}$ too optimistic here?

Exercise B5: Leverage Effect — Romania vs Crypto [Proposed]

Solution discussed in class

Task

- Plot $L(k) = \text{corr}(r_t, |r_{t+k}|)$ for $k = -20, \dots, 20$ on BET-TR and Bitcoin Christie (1982)
- Report $L(1)$ and $L(5)$ for both

Interpretation

- Why is the leverage effect weaker for Bitcoin? Is a correlation enough, or do you need a regression with controls (B12)?

Exercise B6: EUR/RON Across Regimes [Proposed]

Solution discussed in class

Task

- Split EUR/RON log returns into July 2005–2011, 2012–2019, 2020–2026 (the series starts with the new leu, 1 July 2005)
- For each period compute lag-1 autocorrelation, annualised volatility and excess kurtosis

Interpretation

- The fixing is set once a day by the central bank: which statistic would change if you used market quotes instead?

Exercise B7: BET-TR vs S&P 500, Same Period [Proposed]

Solution discussed in class

Task

- ▣ Join the two *price* series on common dates (Sep 2014 – Sep 2026), then take returns
- ▣ CAGR, volatility, Sharpe ($r_f = 0$), Sortino [Sortino & van der Meer \(1991\)](#), maximum drawdown, Calmar
- ▣ Correlation of daily returns on the same day, and with the S&P 500 lagged one day

Interpretation

- ▣ Is the comparison fair (dividends, currency, risk-free rate in RON)? Why is the same-day correlation too low?

Exercise B8: Range-Based Estimators on the S&P 500 [Proposed]

Solution discussed in class

Task

- S&P 500 open, high, low, close from 2008 onwards
- 21-day close-to-close, Parkinson [Parkinson \(1980\)](#), Garman–Klass [Garman & Klass \(1980\)](#), Rogers–Satchell [Rogers & Satchell \(1991\)](#) and Yang–Zhang [Yang & Zhang \(2000\)](#) volatilities

Interpretation

- Yang–Zhang includes the overnight jump, yet stays below close-to-close: which property of the index open explains this (Lecture 1, stale opens)?

Exercise B9: Heteroskedasticity-Robust Ljung–Box [Solved]

Task

- ▣ S&P 500 log returns; $\hat{\tau}_k$ and $\tilde{Q}$ as in the Key Definitions [Lobato, Nankervis & Savin \(2001\)](#)
- ▣ (a) $\hat{\rho}_k$, $k = 1, \dots, 20$, with the classical band $\pm 1.96/\sqrt{T}$ and the robust band $\pm 1.96\sqrt{\hat{\tau}_k}$
- ▣ (b) Classical $Q(10)$ vs $\tilde{Q}(10)$, both against χ^2_{10}
- ▣ (c) Repeat on BET-TR and EUR/RON

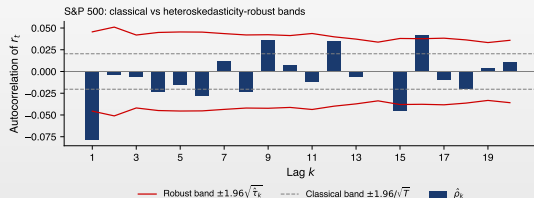
To report

- ▣ $Q(10)$ and $\tilde{Q}(10)$ for the three series; significant lags under each band

Interpretation

- ▣ For which series does autocorrelation survive the correction? Why do BET-TR and EUR/RON behave differently from the S&P 500?
- ▣ What does this say about the Ljung–Box column of the Chapter 1 table?
- ▣ $\hat{\tau}_k$ also assumes a finite fourth moment: what do the Hill estimates say about that?

Solution B9



	S&P 500	BET-TR	EUR/RON
$Q(10)$	91.6	70.0	221.7
$\tilde{Q}(10)$	19.0	16.9	37.4
p of $\tilde{Q}$	0.041	0.076	< 0.001
sig. lags, classical	8	5	6
sig. lags, robust	3	2	2

- All classical $Q(10)$ have $p < 10^{-10}$
- Robust lags: S&P 500 1, 15, 16; BET-TR 2, 10; EUR/RON 1, 3
- Only EUR/RON keeps strong autocorrelation ($\hat{\rho}_1 = 0.17$): the managed fixing, not market inefficiency
- The Chapter 1 Ljung–Box column overstates significance; with $\hat{\alpha} \approx 3$ even $\hat{\tau}_k$ is only approximate

Exercise B10: A Kurtosis That Does Not Converge [Proposed]

Solution discussed in class

Task

- ▣ (a) S&P 500: excess kurtosis on expanding windows, the first n years, $n = 1, 2, \dots, 36$
- ▣ (b) Simulate 500 Student- t series of the same length, with $\nu = 3$ and with $\nu = 6$; compute the same curve and its 5–95% band
- ▣ (c) Overlay the real curve on the two bands
- ▣ Optional: the same for Bitcoin and EUR/RON

Interpretation

- ▣ Which days cause the jumps of the curve? What happens to the kurtosis after each of them?
- ▣ Why does the curve settle for $\nu = 6$ and not for $\nu = 3$?
- ▣ Which band contains the S&P 500? What does “excess kurtosis = 10.9” in the lecture table mean then?
- ▣ Which tail statistic would you report instead of kurtosis?

Exercise B11: Hill on Both Tails, BET-TR and Bitcoin [Solved]

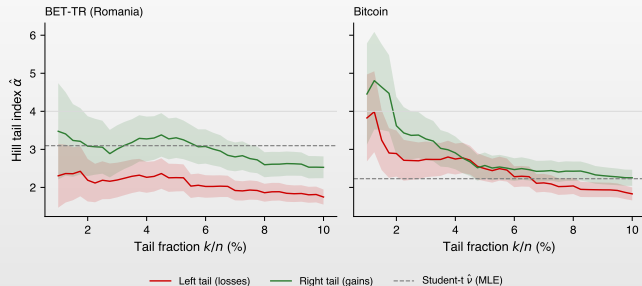
Task

- ▣ (a) Hill plot $\hat{\alpha}(k)$ separately for losses ($-r_t$) and gains (r_t), k from 1% to 10% of n , band $\pm 1.96 \hat{\alpha}/\sqrt{k}$
- ▣ (b) $\hat{\alpha}$ at $k = 2\%$ of n for each tail
- ▣ (c) Approximate asymmetry test:
$$z = (\hat{\alpha}_- - \hat{\alpha}_+)/\sqrt{\hat{\alpha}_-^2/k + \hat{\alpha}_+^2/k}$$
- ▣ (d) Compare with $\hat{\nu} = 3.1$ (BET-TR) and 2.2 (Bitcoin) from B2

Interpretation

- ▣ Does the test confirm the gain/loss asymmetry, or were the quantile differences of the lecture noise?
- ▣ The test assumes independent tail observations: with volatility clustering, is the true standard error larger or smaller?
- ▣ Why does $\hat{\alpha}$ (Hill) differ from $\hat{\nu}$ (Student- t MLE), and which one describes the tail?

Solution B11



- BET-TR ($k = 59$): $\hat{\alpha}_- = 2.19$ [1.63, 2.75], $\hat{\alpha}_+ = 3.08$ [2.30, 3.87], $z = -1.82$ ($p = 0.069$)
- Bitcoin ($k = 87$): $\hat{\alpha}_- = 2.89$ [2.28, 3.50], $\hat{\alpha}_+ = 3.62$ [2.86, 4.38], $z = -1.47$ ($p = 0.14$)
- Losses are heavier in both, but not significant at 5%; clustered extremes make the true SE larger, so the evidence is weaker still
- $\hat{\nu}$ is fitted mostly by the centre: Bitcoin's $\hat{\nu} = 2.23$ lies below both tail indices

Exercise B12: The Leverage Effect as a Regression [Proposed]

Solution discussed in class

Task

- Regression with Newey–West standard errors [Newey & West \(1987\)](#)
 - ▶ $|r_{t+1}| = c + \beta^+ r_t^+ + \beta^- r_t^- + \sum_{j=1}^5 \delta_j |r_{t-j}| + u_{t+1}$
 - ▶ $r_t^+ = \max(r_t, 0)$, $r_t^- = \min(r_t, 0)$
- Symmetric response: $\beta^+ = -\beta^-$; leverage effect: $-\beta^- > \beta^+$
- (a) Test $H_0 : \beta^+ + \beta^- = 0$
- (b) S&P 500, BET-TR, Bitcoin, EUR/RON
- Why is $|r_t|$ not among the controls?

Interpretation

- Which sign do you expect for EUR/RON, and why?
- Why control for $|r_{t-j}|$? What would the regression measure without them?
- Which volatility model of Chapter 5 has exactly this asymmetry?



Exercise B13 (extends B7): Is the Sharpe Difference Real? [Proposed]

Solution discussed in class

Task

- ▣ BET-TR vs S&P 500 on common days: Sharpe 1.27 vs 0.65 (B7)
- ▣ (a) Test of [Jobson & Korkie \(1981\)](#) with the correction of [Mommel \(2003\)](#), on daily Sharpe ratios
 - ▶ $z = (\widehat{SR}_1 - \widehat{SR}_2) / \sqrt{\widehat{V}}$,
 $\widehat{V} = \frac{1}{T} [2(1 - \hat{\rho}) + \frac{1}{2}(\widehat{SR}_1^2 + \widehat{SR}_2^2 - 2\widehat{SR}_1\widehat{SR}_2\hat{\rho}^2)]$
- ▣ (b) Block bootstrap for the difference [Ledoit & Wolf \(2008\)](#)
- ▣ (c) Correlation of BET-TR with the S&P 500 on the same day and on the previous day

Interpretation

- ▣ Is the difference significant? What does the Mommel test assume, and is it plausible here?
- ▣ Bucharest closes hours before New York: why does this matter for the daily correlation?
- ▣ Which correlation would you use in a portfolio [Dimson \(1979\)](#)?
- ▣ Besides dividends and currency, what else makes the comparison unfair (the RON risk-free rate)?



Exercise B14: Which Stylised Facts Does GARCH- t Reproduce? [Proposed]

Solution discussed in class

Task

- ▣ (a) Fit a GARCH(1,1) with Student- t innovations to the S&P 500
- ▣ (b) Simulate 200 paths of the sample length
- ▣ (c) For each: excess kurtosis, ACF of $|r_t|$ at lags 1, 20 and 100, Hill $\hat{\alpha}$, leverage correlation $L(1)$
- ▣ (d) 5–95% bands against the real values: a “reproduced / not reproduced” table

```
from arch import arch_model
am = arch_model(100 * r, vol='GARCH', p=1, q=1,
               dist='t')
res = am.fit(disps='off')
sim = am.simulate(res.params, nobs=len(r))
```

Interpretation

- ▣ Which fact is not reproduced at long lags? Link the answer to the decay $(\alpha + \beta)^k$
- ▣ Can a symmetric GARCH produce the leverage effect? Which change do you propose?
- ▣ For each fact not reproduced, which component of the model should change?

C1: Did BET-TR Autocorrelation Fall after the FTSE Reclassification? [Proposed]

Solution discussed in class

The question

- ▣ BET-TR has positive lag-1 autocorrelation, typical of less liquid markets (Chapter 1)
- ▣ Romania became a FTSE Russell Secondary Emerging market in September 2020
- ▣ Did the autocorrelation fall afterwards? If so, is it a liquidity effect?

Deliverable

- ▣ A Quantlet folder in the team repository and one slide with the conclusion and its main limitation

The answer should contain

- ▣ $\hat{\rho}_1$ on one-year rolling windows, with the robust bands of B9
- ▣ A before/after test with robust standard errors
- ▣ A liquidity measure (volume, number of trades, spread) and its link with $\hat{\rho}_1$
- ▣ Alternative explanations: index composition, new listings (Hidroelectrica, 2023)
- ▣ The link with the Adaptive Markets Hypothesis (Chapter 2)

C2: Critique an AI Answer (1/2) [Proposed]

Solution discussed in class

- **The prompt** a student sent to an AI (Artificial Intelligence) assistant: “Are the daily BET-TR log returns Normally distributed? Are they autocorrelated? Give Python code and a conclusion”
- **The answer:** the code below, followed by the conclusion

```
from scipy import stats
from statsmodels.stats.diagnostic import acorr_ljungbox
r = np.log(bet["close"]).diff().dropna()      # daily log returns
k = stats.kurtosis(r)                          # sample kurtosis
print("excess_kurtosis:", k - 3)               # the Normal has kurtosis 3
jb, p = stats.jarque_bera(r)
lb = acorr_ljungbox(r**2, lags=[10])          # Ljung-Box, 10 lags
```

- **Conclusion of the AI**
 - ▶ “Excess kurtosis 14.8: heavy tails”
 - ▶ “JB p -value 0.000: the probability that BET-TR returns follow the Normal distribution is 0%”
 - ▶ “Ljung-Box $Q(10) = 713.5$, $p < 0.001$: returns are strongly autocorrelated, so BET-TR is predictable and not a random walk”

C2: Critique an AI Answer (2/2) [Proposed]

Solution discussed in class

Task

- (a) Find the errors: there are two or three; the rest of the answer is correct
- (b) For each error, say how you would detect it: a test, a published number, the lecture or the library documentation
- (c) Give the corrected code and the corrected conclusion
- (d) Log each error as in `AI_ERRORS.md`: what was asked, what the AI answered, what was wrong, how you found out, the correct version
- Model: A3, B3 and B9

Why this exercise

- AI code usually runs without an error message: the mistakes are in the statistics, not in the syntax
- Course rule: you may use AI, declare it in `AI_USE.md` and log at least three errors you caught in `AI_ERRORS.md`

Common Pitfalls

Data

- ▣ Mixing price and total-return indices
- ▣ Taking returns before aligning calendars in joint analyses
- ▣ Trusting quote series without checking extremes and autocorrelation
- ▣ Using 252 days for assets that trade every day

Statistics

- ▣ i.i.d. standard errors under volatility clustering
- ▣ Testing a risk model on the sample used to estimate it
- ▣ Reporting a kurtosis when the fourth moment may not exist
- ▣ Comparing Sharpe ratios without a test, or across currencies

- ▣ Rule: look at the data (charts, extremes, zero returns) before running any test

Take-Home Assignment (Team Project)

- ▣ Choose **three** assets from different markets (at least one listed on the Bucharest Stock Exchange)
- ▣ For each asset
 - ▶ Document the data: source, price vs total return, cleaning steps
 - ▶ Summary-statistics table of Lecture 1, with robust autocorrelation tests
 - ▶ Both tail indices with Hill plots; at least five stylised facts of [Cont \(2001\)](#)
- ▣ Compare risk-adjusted performance on a common sample, with a significance test
- ▣ Deliverables: a GitHub repository with a notebook that runs top-to-bottom and a 2-page summary
- ▣ Crypto-assets: read [Härdle, Harvey & Reule \(2020\)](#) before choosing a coin
- ▣ Log every AI error you catch, in this seminar or in the project, in `AI_ERRORS.md`

Further Study on Quantinar

Advanced courses

- ▣  Statistics of Financial Markets
- ▣  Tukey's g- and h- transformations
- ▣  Applied Time Series Analysis with Python

Code

- ▣ **Notebook:** `chapter1_seminar_notebook.ipynb` (Colab)
- ▣ **Quantlets:** github.com/danpele/MFM/tree/main/Quantlets/Ch_01

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